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Lab 8

# Linux Forensics (Artifacts Analysis)

In this lab, we explored Linux forensics tools such as p0f, arp-scan, nmap, swap_digger, mimipenguin, rkhunter, and sleuth kit. These tools can be used to investigate network activity, disk images, and system artifacts. They can help investigators understand information and gather evidence from systems and disk images.

Using p0f, we are able to identify operating systems on a machine by monitoring TCP/IP packets. We can listen to a specific device, list all available interfaces, switch to promiscuous mode, and read packets from a tcpdump. This allows us to gain more information about network characteristics as well.

Using arp-scan and nmap, we are able to search for services and hosts on a network. In the lab, we looked at SYN and UDP scans. These scans are useful because they show us what ports are open, and closed, and which are potentially vulnerable for hackers to use. Using arp-scan is useful to investigators because they provide MAC addresses of devices, and look for devices connected to the network that may have gone undiscovered and are suspicious or pose a threat. Using nmap is useful because it helps to identify all active hosts on a network, even remote ones. Nmap also provides port scanning, and helps to keep track of and maintain networks.

Using swap_digger, we are able to retrieve system usernames, passwords, and other credentials. We were also able to extract other information such as wifi access points, wifi network access list, most accessed resources, files, URLs, and top IP addresses. This is extremely useful in digital forensics because it allows investigators to access those smaller pieces of data that were once in memory, and are now in swap memory.

Using rkhunter and chkrootkit, we are able to scan a system for hidden files, rootkits, and potentially suspicious changes to a system. With these tools, we can look for file signatures using xxd. We can also search files for readable content using ascii or strings.

Using sleuth kit, we used tools such as fls, fsstat, ils, img_stat, img_cat, and fiwalk. These tools help to display general details of a filesystem or image file, output contents, print details, or list files and directories. Using fsstat, we are able to examine filesystem information such as system type, OEM name, root directory, sector size, and cluster size. Using img_stat, we are able to examine image file information such as image type, size, and sector size. This is useful to investigators because they are able to analyze a disk image to gather information about the filesystem, examine metadata, and understand details of files that may have been deleted. This can all be done without the operating system, which makes sleuth kit an extremely valuable tool in digital forensics.

## References

1. Lab 13 pdf